PY4 – Generate prime numbers and set operations

4. Write a Program that generate a set of prime numbers and another set of odd numbers. Display the result of union, intersection, difference and symmetric difference operations.

Aim:

To generate a set of prime numbers and another set of odd numbers, and to display the result of union, intersection, difference and symmetric difference operations.

Coding:

```
odd = set(range (1,10,2))
prime=set()
for i in range(2,10):
        for j in range (2,i):
            if (i%j==0):
                break
            else:
                prime.add(i)
print("Odd Numbers: ", odd)
print("Prime Numbers: ", prime)
print("Union: ", odd.union(prime))
print("Intersection: ", odd.intersection(prime))
print("Difference: ", odd.difference(prime))
print("Symmetric Difference: ", odd.symmetric_difference(prime))
```

Output:

```
Odd Numbers:  {1, 3, 5, 7, 9}
Prime Numbers:  {2, 3, 5, 7}
Union:  {1, 2, 3, 5, 7, 9}
Intersection:  {3, 5, 7}
Difference:  {1, 9}
Symmetric Difference:  {1, 2, 9}
```

Result:

Thus the Python program to generate prime numbers and set operations has been done and the output is verified.

12th Computer Science_EM Practical.indd 337 26-12-2022 17:50:32